What's GPT-∞?

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July 2024

The aim of this piece is to characterize '$\lim_{n\to\infty}$ GPT-n'. In Section 2, we consider a few options for what one might mean by this 'GPT-∞', attempting to make the setup as concrete and simple as possible (so it can be analyzed fruitfully) while maintaining a reasonable degree of relevance to real models. In Section 3 and Section 4, we aim to understand the degree to which each kind of GPT-∞ would be able to do novel science.

The point of understanding GPT-∞ is that this might tell us something about whether language models scale to safe AGI. Section 5 finishes the piece with a discussion of what our claims about GPT-∞ entail about GPT-n.

1 notes

might be easiest to start from the case where data goes to infty, distro of person actually learned. say it still doesn't do that much science!! consider stuff about data density later

issues: definitely need the abili

2 What should we mean by GPT-∞?

We will prioritize simplicity

People have this sense that a dumber verifier might be able to supervise work that goes

3 The model copying the most careful, smart, wise person

crucial question for even perfect 1-day copy: could you solve alignment if your brain is reset after every day? also obviously exactly relevant: HCH

now could be: ok, the damn thing can't learn — this is a problem. but wait, instead of copying a single human, can't i copy all humans? i could even have a training setup where the model gets a bunch of stuff about the life-history of the human, then their question, then has to produce their answer. note this takes out of distro though. also need to see this more clearly in general still why one-day copies: perhaps 1 second or 1 minute or 1 hour would be more appropriate — 1 day will give an upper bound on the capabilities of these versions. the basic issue here is that there much less training data for longer times. A life contains ≈ 8765.82 one-hour chunks for every one-year chunk (this is assuming you don't let the chunks overlap, but it seems reasonable to require disjointness for reasonably independent data points; anyway, if there's a correction here, it would need to be applied to both sides of the equation). but plausibly even more importantly, the one-year distribution is sort of a much bigger object, so we should expect learning it to take potentially many more data points. (i guess the precise dependence has to do with some scaling laws that i can't see clearly atm)

4 The model fine-tuned by a careful smart person

5 Discussion